

ClearTract[®]

FOLEY CATHETER



FDA CLEARED

For Suprapubic, Urethral, and
Nephrostomy Use

THE FIGHT AGAINST CAUTI

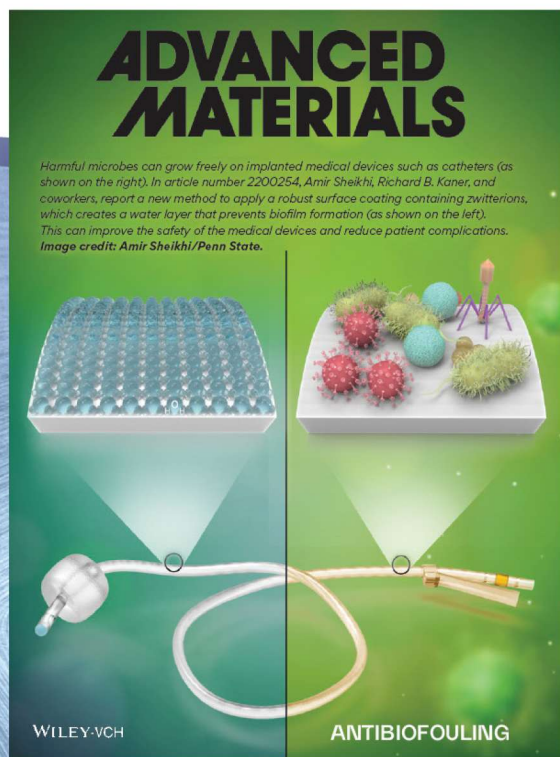
CAUTI is a huge and costly problem worldwide.

- There are now an estimated 1 million catheter-associated urinary tract infection (CAUTI) cases per year... in the United States¹ leading to increased morbidity and mortality for patients, costing hospitals up to \$10,000/patient to treat.²
- “Antimicrobial resistance is an urgent global public health threat, killing at least 1.27 million people worldwide and associated with nearly 5 million deaths in 2019. In the U.S., more than 2.8 million antimicrobial-resistant infections occur each year.” - *Antibiotic Resistance Threats in the United States 2019*, CDC³
- The Society for Healthcare Epidemiology of America advises against the routine use of antimicrobial/antiseptic impregnated catheters for CAUTI prevention.⁴

Silq's groundbreaking zwitterion surface technology

For the past twenty years, scientists have shown that polyzwitterions, a class of bio-inspired polymers, strongly resist the deposition of microbes on surfaces that lead to biofilm formation.¹⁻⁵ Silq has been able to successfully bond polyzwitterions to medical devices at commercial scale, finally unlocking the potential of these polymers. Read more about this in our Advanced Materials article (left).

Silq Technology's proprietary polyzwitterion surface technology was designed to reduce the risk of infection and other complications associated with implanted medical devices⁶. Additionally, Silq's surface treatment process does not rely on the use of antibiotic drugs or silver ions that are designed to kill microbes in the body. These microbes can lead to patient discomfort and the dangerous proliferation of “superbugs” that are resistant to modern antibiotic treatments.⁷⁻⁹



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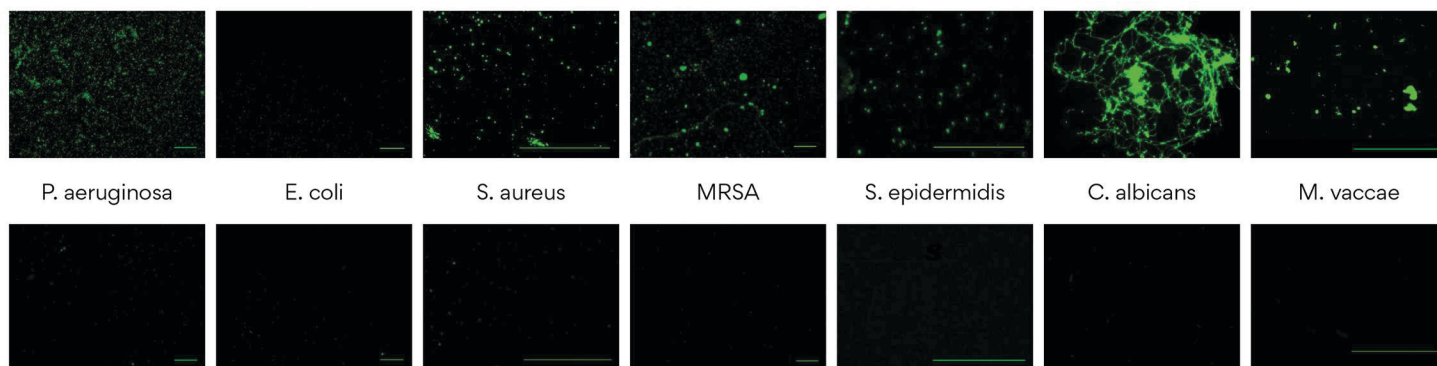


SCAN
HERE
TO
READ

From Adhesion to Colonization

While a catheter is implanted, infectious microbes often anchor to the surface and grow into highly resistant biofilms, reducing the effectiveness of antibiotic treatments for CAUTIs dramatically⁷⁻⁹. Silq's surface treatment aims to hinder this initial adhesion, thereby suppressing CAUTI development before it begins⁶.

SILICONE SURFACE



SILICONE SURFACE WITH SILQ

ClearTract® Clinical Evidence

STC001: Randomized Study to Assess the Ability of ClearTract® Foley Catheter to Reduce Biofilm Formation.

Results of a multi-center, post-market study where 112 subjects were implanted with a randomized catheter type across 8 hospitals in the United States. Upon explant of each catheter, biofilm biomass was quantified by a 3rd party laboratory. (NCT04841226)

CATHETER TYPE	BIOFILM BIOMASS*	SUBJECTS
SILICONE COATED LATEX	289.5	39
SILVER-HYDROGEL LATEX	337.2	38
CLEARTRACT®	82.2	35

*Mass of ATP on explanted catheters (ng/in)

STC002: Long-Term Evaluation of the Silq ClearTract Catheter in Patients With Chronic Urinary Retention. A multi-center, 6-month study comparing the symptomatic CAUTI and encrustation rates of ClearTract® against each subject's current standard of care catheter. Study catheters are replaced monthly, with a crossover at 3 months. (NCT05931887)

CATHETER TYPE	CATHETER EXPLANTED	CATHETER DAYS	UTIs TREATED	AVERAGE ENCRUSTATION
STANDARD OF CARE	226	6729	40 (5.9/1000 Catheter Days)	137.2 mg
CLEARTRACT®	228	7294	13 (1.8/1000 Catheter Days)	38.1 mg

Please note these results are unpublished. Publication under peer review as of 03/03/26

ClearTract™ Catheters: Improves Patient Comfort and Safety

SILQ'S TREATMENT INCREASES CATHETER LUBRICITY BY UP TO 18X,
THEREBY IMPROVING PATIENT COMFORT DURING IMPLANT AND EXPLANT

MADE FROM
100% MEDICAL
GRADE SILICONE

NO
LATEX

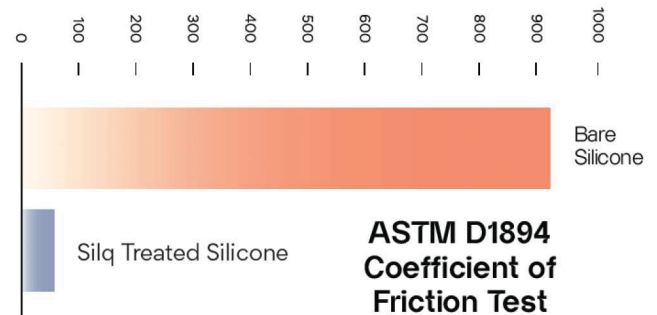
NO
BPA

NO
DEHP

The days of latex healthcare products are coming to an end - 4.3% of the world population has a sensitivity to latex. Healthcare workers have more than double the latex sensitivity rate due to repeated exposure.²⁴ ClearTract™ catheters are made from 100% medical grade silicone, protecting both patients and their healthcare providers.

Regarding safety, ClearTract™ catheters have been cleared by the FDA based on the most recent and more rigorous biocompatibility testing requirements for catheters.

DRAG FORCE (GRAMS)



TESTING SUCCESSFULLY COMPLETED

- Cytotoxicity
- Systemic Toxicity
- Irritation
- Pyrogenicity
- Sub-acute Toxicity
- Sensitivity
- Sub-chronic Toxicity
- Genotoxicity*

*Additional requirement specifically for suprapubic clearance

**All test data available on file

ClearTract® Foley Catheter - Ordering Information

CATHETER SIZE	BALLOON SIZE	VALVE	CATHETER SKU	REFERENCE #
14 Fr	10 cc	2-Way	211410SPT	21400101003
16 Fr	10 cc	2-Way	211610SPT	21600101003
18 Fr	10 cc	2-Way	211810SPT	21800101003

REFERENCES:

1) B Foxman. The epidemiology of urinary tract infection. Nat Rev Urol. 2010; 7(12): 653-660. 2) C S Hollenbeak, et al. The attributable cost of catheter-associated urinary tract infections in the United States: A systematic review. Am J Infect Control. 2018; 46(7): 751-757. 3) CDC. Antibiotic Resistance Threats in the United States, 2019. Atlanta, GA: U.S. Department of Health and Human Services, CDC; 2019. 4) Lo, Evelyn, et al. "Strategies to Prevent Catheter-Associated Urinary Tract Infections in Acute Care Hospitals: 2014 Update." Infection Control & Hospital Epidemiology, vol. 35, no. 5, May 2014, pp. 464-479. 5) Singha, Priyadarshini, et al. "A Review of the Recent Advances in Antimicrobial Coatings for Urinary Catheters." Acta Biomaterialia, vol. 50, Mar. 2017, pp. 20-40. 6) Schleno JB. Zwitterion: coating surfaces with zwitterionic functionality to reduce nonspecific adsorption. Langmuir. 2014 Aug 19;30(32):9625-36. 7) Colilla M, The Role of Zwitterionic Materials in the Fight against Proteins and Bacteria. Medicines. 2018 Nov 22;5(4):125. 8) Kazuhiko Ishihara. Biomimetic materials based on zwitterionic polymers toward human-friendly medical devices, Science and Technology of Advanced Materials, 23:1, 498-524. 9) Liu S Recent Advances in Zwitterionic Hydrogels: Preparation, Property, and Biomedical Application. Gels. 2022; 8(1):46. 10) Yanxian Zhang et al Fundamentals and applications of zwitterionic antifouling polymers. 2019 J. Phys. D: Appl. Phys. 52 403001. 11) Li Q, Zwitterionic Biomaterials. Chem Rev. 2022 Dec 14;122(23):17073-17154. 12) Sin, MC. Hemocompatibility of zwitterionic interfaces and membranes. Polym J 46, 436-443 (2014). 13) Racovita S. Polybetaines in Biomedical Applications. International Journal of Molecular Sciences. 2021; 22(17):9321. 14) Ladd, Jon. Zwitterionic Polymers Exhibiting High Resistance to Nonspecific Protein Adsorption from Human Serum and Plasma. Biomacromolecules (2008). 9. 1357-61. 15) Zhang, L, Cao, Z, Bai, T. et al. Zwitterionic hydrogels implanted in mice resist the foreign-body reaction. Nat Biotechnol 31, 553-556 (2013). 16) B McVerry, et al. A Readily Scalable, Clinically Demonstrated, Antibiofouling Zwitterionic Surface Treatment for Implantable Medical Devices. Adv. Mater. 2022, 34(20): 2200254. 17) Pickard R. Types of urethral catheter for reducing symptomatic urinary tract infections in hospitalised adults requiring short-term catheterisation: multicentre randomized controlled trial and economic evaluation of antimicrobial- and antiseptic impregnated urethral catheters. Health Technol Assess 2012;16(47). 18) McNeilly, Oliver, et al. "Emerging Concern for Silver Nanoparticle Resistance in Acinetobacter Baumannii and Other Bacteria." Frontiers in Microbiology, vol. 12, 16 Apr. 2021. 19) Hosny, Alaa El-Dien MS, et al. "The Increasing Threat of Silver-Resistance in Clinical Isolates from Wounds and Burns." Infection and Drug Resistance, vol. Volume 12, July 2019, pp. 1985-2001. 20) Köves B, Magyar A, Tenke P. Spectrum and antibiotic resistance of catheter-associated urinary tract infections. GMS Infect Dis. 2017 06. 21) W Xu, et al. Host and bacterial proteases influence biofilm formation and virulence in a murine model of enterococcal catheter-associated urinary tract infection. Biofilms Microbiomes. 2017; 3: 28. 22) D Sharma, et al. Antibiotics versus biofilm: an emerging battleground in microbial communities. Antimicrob Resist Infect Control. 2019; 8: 76. 23) M Wu, et al. Current prevalence rate of latex allergy: Why it remains a problem?. J Occup Health. 2016, 58(2):138-144.